

What is the black hole information problem about?

Effective-theory versus full-theory purity

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Abstract

The black hole information problem is usually given as: is the final post-evaporation black hole exterior state pure or mixed? Or, if we follow the Page-time version, how and when does the black hole exterior stop generating entanglement entropy, so as to remain consistent with the thermodynamic description provided by the Bekenstein-Hawking entropy upper bound? We argue that the typical expositions of these questions leave ambiguous the distinction between purity in the full quantum gravity theory and purity in the effective theory, and that the two need not agree. This distinction matters, since if unitarity holds on the effective side, the no-drama equivalence principle is incompatible with the black hole exterior's eventual purification in the effective theory, regardless of whether the full theory is itself unitary and reaches a pure final state. At least one of three things has to be given up in the effective theory: the equivalence principle, effective bulk Hilbert space factorization, or unitarity of the effective theory. The answers offered by mainstream approaches are unclear, though they often come close to implicitly giving up effective-theory unitarity without acknowledgment - which very likely amounts to either new non-linear effective physics or a new understanding of quantum measurement, whose acceptability to physicists also remains unclear.

1 Introduction

In this paper, we explore what the contemporary black hole information problem really is about, updating the progress made in Belot, Earman and Ruetsche (1999) [1] and Wallace (2018) [2]. We assume that the intermediate-time (Page-time) black hole information problem (or paradox) is a valid problem, though this can be contested.

The key thesis of this paper (**‘primary thesis’**) is that the Page-time paradox is about both the quantum gravity theory (or ‘fine-grained theory’) **and** the effective theory (or ‘coarse-grained theory’). This is a serious issue, because most mainstream quantum gravity physicists nowadays almost only tackle the quantum gravity theory part of the information problem as if that resolves the entire information problem. We discuss the mainstream intuitions behind why the Page-time paradox is often considered to be purely about the quantum gravity theory throughout this paper - mostly around the ensemble approaches to the effective theory, Haar-random intuitions from the Page theorem, Ryu-Takayanagi (RT) generalizations - and show why none of them means the effective theory can be safely ignored - the main statement is made in Section 2.3, but we explore this issue throughout the paper.

That the Page-time paradox is also about the effective theory mainly has to do with the three things:

1. Thermodynamic descriptions of black holes are supposed to be the most coarse ones even within the effective theory. If the effective theory produces the black hole entropy that exceeds the thermodynamic Bekenstein-Hawking entropy, then we should search where the effective theory over-counts. Resolving this then involves the effective theory.
2. There is no theory of quantum gravity that currently passes sufficient empirical checks. Therefore, whether the quantum gravity theory produces the expected effective-theory results is important, especially since our observations, at least for the time being, are done with effective-theory observables. In particular, whether the quantum gravity theory actually satisfies the ‘no-drama’ equivalence principle around the black hole horizon is an important question.
3. Around the final evaporation, we are supposed to believe that the black hole interior practically disappears, and this is supposed to be visible in the effective theory as well. Even if the quantum gravity theory produces the close-to-zero entanglement entropy, it may still be possible that the assumed effective theory has an extremely large entropy, implying some issues in coarse-graining or in the full quantum gravity theory itself.

We then proceed to the question related to the **‘secondary thesis’**: can the effective-theory Page-time paradox be resolved by the current proposed resolutions of the quantum-gravity Page-time paradox? We factor in the limitations posed by many mainstream resolutions relying on holographic gravity and mostly assume that holography is not an issue. Exploring the question

is largely done in terms of the generalization of the small-corrections theorem in Section 3. The intuition behind the small corrections theorem [3], especially its generalization, is simple: if the *corrected* effective theory, after receiving corrections from quantum gravity, is unitary, then exponentially small corrections to the uncorrected effective field theory over small time interval can only accumulate to very small corrections for a non-exponential time interval.

Therefore, the ‘**secondary thesis**’ of this paper states that the effective-theory Page-time paradox can only be resolved by giving up at least one of the three (though we can limit the degree of the breakdown as seen in Section 2.12):

1. No-drama equivalence principle around the black hole horizon. (Giving up the equivalence principle suggests that the breakdown of general relativity is large at late evaporation even for a small time interval.)
2. The global Hilbert space factorizing into the black hole interior and the exterior in the effective theory. (This makes the Page-time paradox trivial, as there is no entanglement entropy to define.)
3. The unitarity of the effective theory. (A non-unitary effective theory can nevertheless be consistent with a unitary S-matrix despite intermediate-time non-unitarity.)

While we restrict to the Page-time ‘entropy’ paradox, the black hole information problem, or more precisely its generalization, is richer in quantum gravity especially for AdS/CFT holography. There, the question involves different consistency issues, not just entropy - see [4] for more details.

2 What is the black hole information problem about?

2.1 Belot, Earman and Ruetsche (1999): the final state purity information problem is not a genuine paradox

In philosophy of physics, it is known that the black hole information problem may not be well-posed. Belot, Earman, and Ruetsche (BER) [1] construct an algebraic quantum field theory argument [5] that the generic black hole evaporation spacetime, especially for actual astronomical black holes, carries an initially pure state to a final mixed one, on the grounds that unitarity is guaranteed only across a transition between two Cauchy slices.

2.2 Wallace (2018): the mainstream black hole information problem is about the Page-time (intermediate-time) and thermodynamic consistency paradox, not the final state purity paradox

A relatively recent counter to the BER discussion is by David Wallace [2], where he distinguishes between the evaporation-time paradox (which is covered by the BER discussion) and the Page-time paradox. Wallace argues that mainstream quantum gravity physicists often mean the Page-time paradox when they refer to the black hole information problem. This is confirmed by quantum gravity physicist Suvrat Raju [4], who shows that the final purity paradox is not difficult to resolve, once we know that a typical mixed state can easily be resolved (or ‘approximated’) to a pure state when available observables and operators are constrained to be simple - a more conventional approach invokes the eigenstate thermalization hypothesis. As Raju then states, the contemporary paradox is therefore about the intermediate-time conflict between the effective field theory view and the thermodynamic view of an evaporating black hole, especially around the time at what we call as Page time [6]. See also the discussion by string theorist Samir Mathur [3], which also presents the small corrections theorem in light of this intermediate-time paradox.

The Page-time paradox is simple to state, visually represented in Figure 1. Quantum gravity physicists largely adhere to the black hole thermodynamics view (derived initially independent of quantum physics), which suggests that coarse-grained thermodynamic entropy of a black hole follows the first law of thermodynamics with the black hole mass M considered to be dU and $\frac{\kappa}{8\pi}dA$ (with A being the black hole horizon area) analogous to TdS - in the natural unit:

$$dM = \frac{\kappa}{8\pi} dA + \Omega dJ + \Phi dQ, \quad (1)$$

The later discovery of Hawking radiations in quantum field theory sets black hole temperature $T = \kappa/2\pi$, with thermodynamic entropy S_{th} :

$$S_{th} = \frac{A}{4} \quad (2)$$

The core issue is that while black hole temperature T and thermodynamic entropy are made precise by quantum field theory, this is largely a classical result that should set the upper bound to the entanglement entropy, but as we see in Figure 1, the intermediate-time collision is

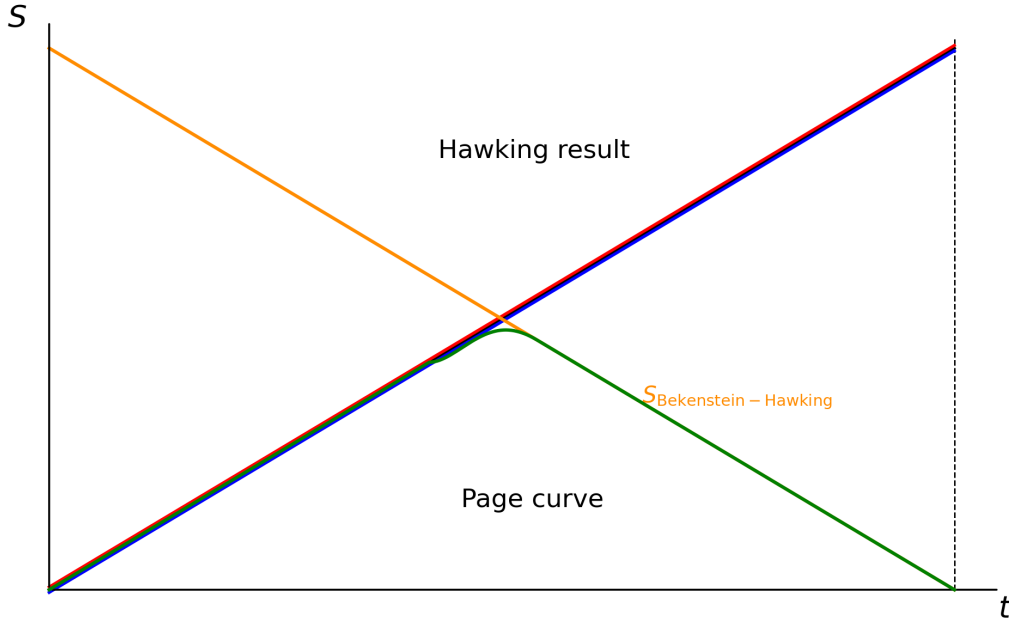


Figure 1: The Bekenstein-Hawking entropy, represented by the monotonically decreasing yellow line, is an approximate coarse-grained entropy upper bound to entanglement entropy. The Hawking entropy result is a purple monotonically increasing line, predicted by semiclassical effective field theory. The problem is that the semiclassical entropy result (purple line) goes above the Bekenstein-Hawking entropy (yellow line) around the Page time, violating the upper bound. The ideal Page curve is given by the green curve.

inevitable, as the effective field theory calculation predicts a monotonic increase in entanglement entropy. Thus in the contemporary paradox, it is not enough to simply argue by final state purity or non-purity - we have to either resolve what happens in the middle of evaporation or deny the Bekenstein-Hawking entropy being an upper bound to entanglement entropy.

2.3 What is the entropy upper bound actually bounding? Coarse-grained entropy of the effective field theory or fine-grained entropy of the full quantum gravity theory?

But at this point, we are led to the question of what this entanglement entropy upper bound actually refers to. Is it bounding the entanglement entropy of the effective field theory on black hole spacetime? Or is it bounding the fine-grained entanglement entropy?

The key issue is that the thermodynamic entropy (Bekenstein-Hawking entropy) is supposed to be the ‘worst’ coarse-grained entropy possible. So if the effective theory is producing the entanglement entropy much more than this worst-case scenario, then we have to work out what the effective theory further misses (or misleadingly further includes).

Even if the quantum gravity theory turns out to respect the thermodynamic entropy upper

bound, what goes wrong on the effective theory is important, right because we have no guarantee that our understanding of the quantum gravity theory is correct, consistent with the empirical (and effective) reality. In this case, the main issue is the ‘no-drama’ **equivalence principle**: that around the black hole horizon, nothing surprising can be visible and the general-relativistic smooth horizon along with the effective field theory calculations remains correct, requiring only small corrections. It can turn out that even with corrections from the assumed theory of quantum gravity, the effective-theory entanglement entropy increases far beyond the thermodynamic entropy.

Therefore, the black hole information problem is partially about the coarse-grained entropy of the effective theory. Even if the coarse-grained version is resolved, this does not mean that the full black hole information problem is resolved - the entanglement entropy upper bound set by the thermodynamic entropy still binds the fine-grained entanglement entropy of the full quantum gravity theory. Nevertheless, most of the conceptual issues regarding the black hole information problem arise right because the coarse-grained entropy issue is ignored - in fact, it is relatively easy to find the proposals demonstrating how the fine-grained entropy respects the thermodynamic-entropy upper bound, and therefore the focus of this paper is on the coarse-grained entropy.

2.3.1 Final evaporation problem for the effective theory

The most serious reason why the thermodynamic upper bound issue is also critical for the effective theory has to do with the final black hole evaporation.

It is true that in light of epistemic ignorance, a pure state may sometimes be approximated as a mixed state. For example, I know that a fair coin dropped from my table would have landed with either head or tail (but not its mixture or superposition) before I actually observe it, but before the observation, I can only assign the probability of $1/2$ to each outcome. In such a case, I would represent the coin state in terms of a mixed state. However, the whole point of corrections from quantum gravity on the effective theory is to fill in this epistemic ignorance on the effective theory.

In case of the final evaporation, the black hole interior Hilbert space dimension shrinks close to zero according to the Bekenstein-Hawking entropy even within quantum gravity. This means that the black hole exterior is very close to being a pure state, and it would be this black hole exterior that we would be coarse-graining to the effective theory. It is difficult to see how an

additional relation between the interior and the exterior would arise just from coarse-graining. The problem is not just minor, as the effective field theory predicts that the entanglement increases monotonically until the final evaporation, very far away from no entanglement.

Indeed, the ‘opposite’ case is more likely - because the effective theory cannot extract entanglements within the full quantum gravity theory, a mixed state may actually be viewed as a pure state in the effective theory, especially in light of the quantum error-correcting code proposal to holographic gravity.

2.3.2 The possibility of an effective field theory being an ensemble theory such that the thermodynamic upper bound does not bind the effective theory

It is true that in recent years, effective field theories are sometimes viewed as an ensemble theory. In this case, an effective field theory (EFT) only really predicts an average state possible for an ensemble of possible theories. In most cases, the self-averaging property implies that this average state is the actual effective state, but whenever self-averaging breaks down, an average state does not fully represent an actual state.

Under this view, EFT is not a usual coarse-grained version of the full quantum gravity theory. Rather, it reflects effective-theory uncertainty over quantum gravity. The semiclassical spacetime of the EFT, along with the equivalence principle, therefore reflects this averaged picture.

Because the EFT reflects an ensemble, there is nothing wrong under the ensemble view with its prediction of the monotonic increase in the entanglement entropy between the black hole interior and the exterior. What should be checked is instead the average of the actual entanglement entropy, not the entropy of the average state.

As reviewed in Section 2.4, the Page theorem [7] states that a typical Haar-random state respects the Page curve such that the average entanglement entropy is reduced to zero at the final evaporation. If we take the ensemble to reflect quantum gravity’s Haar-randomness from the point of the effective theory, then we have the following answer to the upper bound question. The Bekenstein-Hawking thermodynamic upper bound constrains only quantum gravity because the effective field theory is an averaged version (dynamics of average states).

In holographic gravity, this is often realized in terms of the JT/RMT duality, where JT gravity (treated as an ensemble theory) on the bulk spacetime with replica wormhole (and in general, supposedly non-perturbative) corrections is dual to a random matrix theory on the

boundary. Saad, Shenker, and Stanford ‘demonstrated’ this duality in the double-scaled limit: the disk-level JT gravity path integral is reproduced by the matrix integral

$$Z(\beta) = \int dH e^{-L \text{Tr } V(H)} \text{Tr } e^{-\beta H}, \quad (3)$$

over Hermitian $L \times L$ matrices H , with V tuned so that the resulting eigenvalue density reproduces the Schwarzian density of states [8]. This duality is then extended to higher-level path integrals such as computing $\langle Z(\beta_1)Z(\beta_2) \rangle$, but this requires that we take the previously discarded contributions seriously - more specifically, higher-genus JT wormhole geometries connecting n asymptotic boundaries. Naively and naturally read, these contributions represent average ensemble properties that averaged states alone cannot provide. (However, we note that the non-ensemble interpretation of these replica wormhole contributions as ‘non-perturbative corrections to semiclassical gravity from quantum gravity’ is more popular for whatever reasons.)

However, the ensemble view does not state that usual coarse-graining is impossible. Therefore, we can still ask what happens for the usual coarse-graining of a quantum gravity theory into an effective theory. This similarly applies for the computational complexity arguments, which say that observer limitations allow for the resolution of the information problem (the BER paper also discusses this briefly). In terms of the ensemble view, we ask what the actual effective theory behaves like, instead of the full ensemble effective theory. What really needs to be broken in the ensemble theory to resolve the Page-time paradox beyond the reasoning by randomness averaging? It turns out that this question cannot be answered so easily *even if* we take the ensemble view to be the obvious truth, taking the cues from the Page theorem.

The question of firewalls within JT gravity with replica wormholes further complicate the matter [9, 10], and the ensemble view is not as straightforward as it initially sounds to be, and likely requires the complementary non-ensemble analysis.

2.4 A brief review of the Page theorem that underlies the Page time, and why the black hole information problem is a dynamical problem, not a kinematic problem

We briefly go over the Page theorem [7] that is relevant for the concept of the Page time [6]. If a bipartite quantum system of total Hilbert space dimension $d_{BH}d_{rad}$ is in a Haar-random pure

state, the average entanglement entropy of the smaller factor, of dimension $d_{rad} \leq d_{BH}$, is

$$\langle S_{rad} \rangle = \sum_{k=d_{BH}+1}^{d_{BH}d_{rad}} \frac{1}{k} - \frac{d_{rad}-1}{2d_{BH}} \simeq \ln d_{rad} - \frac{d_{rad}}{2d_{BH}}, \quad 1 \ll d_{rad} \leq d_{BH}, \quad (4)$$

so that a typical bipartite state is nearly maximally entangled. Applied to black hole evaporation, with $d_{rad}(t)$ and $d_{BH}(t)$ the dimensions of the radiation and remaining-black-hole factors at observer time t , Page time is the point at which $d_{rad}(t) = d_{BH}(t)$.

Before the Page time, Equation (4) tracks the semiclassical, monotonically increasing curve of Figure 1. After the Page time, $\langle S_{rad} \rangle$ instead tracks $\ln d_{BH}(t)$, because the inequality between d_{BH} and d_{rad} has flipped.

Technically, this Page curve kinematically settles the fine-grained entropy question of the black hole information problem. But there is still a dynamical issue - how and why do dynamics of some fine-grained theory and its derived effective theory support this Page curve? That is, how does the actual evolution work out to remain in these typical states? It could be that evaporating black hole states are not typical within the entire Hilbert space, and problematic calculations that we partially explore in this paper increase the possibility of atypicality. This dynamic evolution part remains unsettled and continues to haunt modern approaches that use typicality to resolve the black hole information problem.

2.5 Quantum gravity physicists typically assume that the information problem is about fine-grained entropy - AdS/CFT view

Quantum gravity physicists typically and implicitly assume that the upper bound is approximately bounding the fine-grained entanglement entropy and do not care much about the effective theory. In one part, this is due to the Ryu-Takayanagi (RT) formula in the AdS/CFT correspondence (where the bulk AdS theory is incomplete and only effective such that can non-perturbatively be completed by CFT), which has the same area law as Equation (2). The RT formula states that the entanglement entropy of some boundary sub-region A in the conformal field theory (CFT) can be calculated as the area of the extremal surface γ_A homologous to A in the holographically dual AdS bulk,

$$S(A) = \frac{\text{Area}(\gamma_A)}{4G_N}, \quad (5)$$

minimized over all such extremal surfaces [11]. This has been extended several ways, including the recent islands proposal that is relevant for the black hole information problem.

We briefly note the islands proposal: the islands proposal computes a generalized entropy for the radiation R ,

$$S(R) = \min_X \text{ext}_X \left[\frac{\text{Area}(X)}{4G_N} + S_{\text{semicl}}(R \cup I_X) \right], \quad (6)$$

extremizing over candidate quantum extremal surfaces X , with any resulting island region I_X behind the horizon included in the entropy calculation [12, 13]. We also note that the islands proposal remains somewhat controversial even within holographic gravity, including a powerful critique that graviton has to be massive in a quantum gravity theory with islands [14], though there are some complications which include the question of Hilbert space factorization.

Since the RT formula tackles the fine-grained entropy, the black hole information problem is often automatically cast in terms of the fine-grained entropy of the full quantum gravity theory.

However, the problem is that the Page-time paradox partially involves the question of what goes wrong in the effective (bulk) field theory around the Page time. Within the AdS/CFT correspondence, this means we have to look at what happens in the effective AdS bulk field theory. In principle, this seems straightforward - if the direct bulk field theory calculation is difficult, we can take it to the holographically dual CFT. But in practice, details of AdS/CFT are far from being completely understood and refinements in our understandings are being constantly made [15]. Furthermore, notwithstanding the complications inherent to the anti-de-Sitter (AdS) space that prevent large AdS black holes from evaporating, the islands proposal, at least within the natural and the naive interpretation, suggests that the Hilbert space of the exterior radiations actually covers some parts of the black hole interior. This is sometimes defended through the idea that non-perturbative corrections, which are step-wise small, eventually accumulate to large corrections such that the naive AdS bulk field theory is wrong - this is sometimes done through the replica wormholes proposal [16, 13].

2.5.1 Replica wormholes do (not) address the coarse-grained effective bulk entropy problem

The replica wormholes proposal also supposedly addresses where the effective bulk calculation goes wrong by the addition of a new hidden saddle in effective bulk path integrals that becomes

dominant around the Page time, replacing the existing semiclassical saddle.

The proposal first notes that the Rényi entropies of the radiation can be computed via the replica trick. Preparing n copies of the Euclidean path integral for the state and gluing them cyclically along R , the unnormalized quantity $\text{Tr } \rho_R^n$ (ρ_R being density matrix) can be computed as a gravitational path integral Z_n evaluated on the n -sheeted replicated boundary geometry:

$$\text{Tr } \rho_R^n = \frac{Z_n}{Z_1^n}, \quad (7)$$

with Z_1 the ordinary, single-copy Euclidean path integral and,

$$Z_n = \int_{\partial g = \mathcal{M}_n} \mathcal{D}g \, e^{-I_{\text{grav}}[g]}, \quad (8)$$

When Z_n is defined this way, it includes all geometries not included for the purpose of using the replica trick, and it is where the main controversy regarding replica wormholes arises. From the proponents point of view, the proposal is extremely intuitive, with conceptual connections to string theory that provide the justification of new geometries providing non-perturbative quantum-gravity corrections to the effective theory.

We should distinguish the replica trick within the quantum gravity theory and the replica trick with the effective theory - the proponents of the replica wormhole proposal are only discussing the replica trick applied to the effective theory, not the full quantum gravity theory. Therefore, the argument that the replica trick used this way is simply mathematically wrong is less persuasive than it initially sounds. But we now see the hint of the problem we explore throughout the paper - is the effective theory unitary? The corrections to the Euclidean theory clearly relate to the corrections to the Lorentzian theory, and such corrections to quantum mechanics usually imply non-unitarity.

The replica trick for the Rényi entropies says:

$$S_n(R) = \frac{1}{1-n} \ln \text{Tr } \rho_R^n \quad (9)$$

The entanglement entropy is recovered from the Rényi entropies as:

$$S(R) = \lim_{n \rightarrow 1} S_n(R) \quad (10)$$

When Z_n from Equation (8) is used, this produces the desired Page curve [17, 18].

Therefore, we could say that the replica wormholes approach also addresses both the fine-grained entropy and the coarse-grained entropy version of the information problem. However, as foreshadowed in the non-unitarity concerns arising from corrections to the replica trick, it turns out that the matter is not so simple. In terms of Equation (21) which is central to this paper and is a general version of the small corrections theorem [3], the replica wormhole proposal corresponds to the effective field theory (EFT) calculation itself already requiring corrections with no monotonic increase in the entanglement entropy on the corrected effective field theory as well. (The baseline effective theory then refers to the corrected effective field theory.)

But Equation (21) is set up such that the baseline effective theory can be non-unitary (which means we can equate the baseline effective theory/EFT to the uncorrected EFT), only requiring the unitarity of the entire corrected effective theory (which may not even be a field theory). We can then ask whether incremental corrections via sub-dominant saddle contributions (‘replica wormhole’) can accumulate to large corrections around the Page time - they cannot.

Replica wormholes therefore need to give up some of the conventional assumptions, which we identify as the effective bulk Hilbert space factorization, unitarity of the effective theory or the small corrections to the equivalence principle. The last sacrifice is unavailable, since replica wormholes assume from the start that the corrections are small for small time intervals, so what remains are the first two options.

2.6 The equivalence principle question for the bulk effective field theory: can the unitarity of the quantum gravity theory + islands really address the information problem without involving the effective bulk field theory?

The AdS/CFT + islands argument often presupposes that the bulk effective field theory discussion is unnecessary in holography - after all, if we know that the bulk has to be non-perturbatively corrected and the correct quantum gravity description can be used, why even reduce to the effective field theory and not use holographic machinery?

There are two problems with this stance. The first one is that this AdS/CFT model can give the wrong effective bulk that is wildly inconsistent with the actual black holes to the point that the model has no value even as a toy model. In particular, the no-drama equivalence principle may be significantly violated such that local observers notice something significantly different from general relativity and the effective field theory around the black hole horizon. The second

one is that even if the equivalence principle is not significantly violated (thereby respecting the ‘small corrections’), the resulting effective bulk field theory can be non-unitary.

2.7 The unitarity question for the effective bulk field theory: distinguishing effective-theory unitarity from quantum-gravity unitarity

Even if the current understanding of AdS/CFT gives the right effective bulk, we have to ask whether this is a unitary development in the effective theory. This requires distinguishing effective (bulk) unitarity and (boundary CFT) unitarity within the quantum gravity theory.

It could be argued that because the full quantum gravity theory is unitary, the ideas like replica wormholes, or ‘small non-perturbative corrections accumulating to develop a new saddle in path integrals,’ are automatically unitary in the effective bulk field theory, at least the non-perturbatively corrected ones. But this is something to be verified - it could be that we have misunderstood AdS/CFT, or the islands formula itself may be false, and any demonstration of non-unitarity would require a change in the understanding.

The unitarity of the effective bulk field theory is a serious matter, because we pretty much assume it implicitly everywhere. Our quantum measurements are made in terms of effective bulk observables, not quantum gravity observables. Our quantum computing all implicitly assumes effective unitarity, not just the unitarity of the full quantum gravity theory. The non-unitarity of the effective bulk field theory therefore implies either a new effective physics or a new theory of quantum measurements, none of which can be taken and accepted lightly.

2.8 Which to sacrifice: Hilbert space factorization, unitarity or the equivalence principle in the effective bulk field theory?

The non-unitarity of the effective bulk field theory, derived in the small corrections theorem [3] and to be derived more generally in this paper (Equation (21)), assumes the small corrections to the no-drama equivalence principle and the ability to factorize the global Hilbert space into the black hole exterior and the interior subspace. This implies that the equivalence principle or the Hilbert space factorization can be denied instead. We do not go in detail regarding sacrificing the equivalence principle, other than to note that a major heterodox but string-theoretic position in this regard is the fuzzball argument [19], which rejects the existence of smooth black hole horizons given as in general relativity.

2.9 Islands/quantum error-correcting codes view: sacrificing effective bulk Hilbert space factorization?

Sacrificing the Hilbert space factorization is hinted by the naive and natural interpretation of the islands proposal, a mainstream quantum gravity position.

The islands proposal says that the entanglement wedge of black hole exterior radiations must cover the black hole interior as well [12, 13]. This is best understood within AdS/CFT, where black hole exterior radiations are defined on the boundary CFT and we ask how they are represented on the AdS bulk.

The problem is that simply re-defining the black hole interior and the exterior will not work to avoid the non-unitarity problem. Therefore, any islands proposal respecting a unitary effective theory requires an eventual complete breakdown in the ability to factorize the global Hilbert space into the black hole interior and the exterior. With this breakdown, the Page-time paradox on the effective bulk field theory is resolved trivially - there is no black hole interior to define the entropy on. What we call as the Page curve only makes sense on the holographic boundary - that is, only on the full quantum gravity theory, not the effective theory. This also captures the idea that the holographic boundary theory redundantly encodes the bulk in terms of quantum error-correcting codes, which is why the bulk Hilbert space factorization eventually breaks down [20].

A minimal illustration of this redundancy is the three-qutrit code - a single logical qutrit $|\psi\rangle_L = \alpha|0\rangle_L + \beta|1\rangle_L + \gamma|2\rangle_L$ is mapped to three physical qutrits via $|j\rangle_L \mapsto \frac{1}{\sqrt{3}} \sum_{k=0}^2 |k, k+j, k+2j\rangle \pmod{3}$, such that erasing any single physical qutrit still leaves the logical state fully recoverable from the remaining two. The bulk-to-boundary map in this error-correcting code holography behaves similarly - a bulk operator deep in the interior is redundantly represented within many different boundary sub-regions. See the Hayden-Preskill result for an early error-correcting code holography prototype [21].

2.10 The possibility of non-unitarity effective theory

A non-unitary theory, including a non-linear effective theory is not really absurd - as noted in the BER argument and Maudlin [22], the Page-time paradox involves transitions that are not from a Cauchy surface to another Cauchy surface. Maudlin [22] further notes that even within the AdS/CFT correspondence, the boundary theory exhibiting unitarity does not imply

the bulk unitarity in case the boundary respects the Cauchy-to-Cauchy transition but the dual bulk theory does not. Therefore, intermediate unitarity should not be expected, and resolving the Page-time paradox only imposes a unitary S-matrix between the initial and the final times due to the thermodynamic entropy going to zero.

2.10.1 There may be a unitary S-matrix even under the non-unitary effective bulk theory

If we insist on no final loss of information on the effective theory, information has to be restored in the effective theory. More technically, this means that there is a unitary S-matrix for black hole evaporation despite intermediate non-unitarity.

Further assume that the effective field theory is approximately followed for early evaporation. The Page-time paradox is then resolved as follows, requiring no breakdown in the Hilbert space factorization.

For early evaporation, the effective theory remains unitary, largely identical to the effective field theory. Then around the Page time, the effective theory becomes non-unitary and quantum gravity starts to provide corrections such that information is not lost in the effective theory after the final evaporation. In this case, while the effective theory allows for intermediate non-unitarity, there is a unitary S-matrix for black hole evaporation.

2.10.2 A feature of non-linear effective theory: different effective states evolving to the same effective state

One feature of such a non-linear theory is that different effective states can evolve to the same effective state. This is very simple to understand - unitary evolution (by unitary U) respects orthogonal states $\langle\psi_a|\psi_b\rangle = 0 \Rightarrow \langle\psi_a|U^\dagger U|\psi_b\rangle = 0$ but if evolution is by some non-unitary operator O_{NU} then $\langle\psi_a|\psi_b\rangle = 0 \Rightarrow \langle\psi_a|O_{NU}^\dagger O_{NU}|\psi_b\rangle \neq 0$ is possible.

2.11 The possibility of no effective theory covering both the black hole interior and the exterior at late times

Alternatively, the effective theory of both the black hole interior and the exterior may not exist at late times, which becomes the reason why the Page-time paradox cannot exist in both the effective theory and the quantum gravity theory.

The most radical view in this regard is holography of information, which suggests that in quantum gravity, the black hole exterior already has all the information about the black hole interior [23]. This has to do with the gravitational Gauss law, which we do not discuss in depth in this paper. Because quantum gravity only requires the black hole exterior, its effective theory should only require the black hole exterior as well. The conventional field equations respecting the equivalence principle with small corrections may be *reconstructed*, but its Hilbert space only requires the black hole exterior.

Holography of information typically argues by the following intuition: in any theory with a gauged long-range symmetry, charges cannot be created or destroyed locally - the total charge Q enclosed by a closed surface $\partial\Sigma$ is fixed by the flux through $\partial\Sigma$:

$$Q = \oint_{\partial\Sigma} \star F, \quad (11)$$

and gravity behaves the same way, with the ADM/Komar mass and momentum likewise given by surface integrals evaluated arbitrarily far from any source. Any operator that creates an excitation deep in the black hole interior must therefore be *dressed* with a long-range gravitational (or gauge) tail reaching all the way to the boundary of the region. This dressing implies that within a theory of quantum gravity, the black hole exterior already has a record of the interior excitation [23].

2.12 An effective quantum gravity understanding based on conservation of energy

So far the arguments went as if the Hilbert space factorization breakdown or the equivalence principle breakdown has to be ‘wholesale’ at each time. This does not have to be the case, right because we already know that at each time, there is approximately a superposition of spacetime, having ignored additional pure gravity effects in quantum gravity.

2.12.1 Approach 1

Because Hawking radiations are thermal, conservation of energy requires that spacetime is in superposition. Take some spacetime branches that do not have much black hole mass left. If we follow the view that the effective Hilbert space factorization breaks down, there is no need to break the factorization on all branches of the spacetime superposition - instead, we only consider

the branches with minimal black hole mass to have the black hole interior that is inaccessible, whether due to its actual non-existence or simply its inaccessibility. Then the effective universe state becomes something like:

$$|\Psi_t\rangle = \sum_{k \notin \mathcal{K}_t} \alpha_{k,t} |\psi_k\rangle + \sum_{k \in \mathcal{K}_t} \beta_{k,t} |\psi'_k\rangle, \quad \sum_k |\alpha_{k,t}|^2 + |\beta_{k,t}|^2 = 1 \quad (12)$$

where $\mathcal{K}_t$ refers to the set of spacetime that does not admit the black hole interior accessibility at time t . In this case, even if there are only exponentially small corrections to the equivalence principle on the branches $k \notin \mathcal{K}_t$, over time these branches evolve to branches $k \in \mathcal{K}_t$ such that $\sum_{k \in \mathcal{K}_t} |\beta_{k,t}|^2$ continues to monotonically increase.

If we assume that $|\psi'_k\rangle$ (with $k \in \mathcal{K}_t$) would then be approximated as a pure effective state with support purely within the black hole exterior, then the Page-time paradox is resolved in the effective theory, only requiring the effective Hilbert space factorization breakdown in the branches with minimal black hole mass. The quantification of what counts as minimal black hole mass at each time is provided by the Page curve calculation. In this case, the Page curve becomes a dynamic theory that determines which spacetime branch would have the Hilbert space factorization breakdown - that is, the Page curve becomes the theory determining each $\mathcal{K}_t$. The final state purity only requires that $\sum_{k \in \mathcal{K}_t} |\beta_{k,t}|^2$ keep monotonically increasing over time up to 1, and does not require the Page curve constraint.

The transitions from branches $k \notin \mathcal{K}_t$ to $k \in \mathcal{K}_t$ can certainly be unitary in this case - intermediate non-unitarity only arises because $k \in \mathcal{K}_t$ branches do not have the accessible black hole interior and therefore are approximated by the pure effective black hole exterior state. Whether a unitary S-matrix is established depends on pure-state quantum-gravity corrections to the effective theory, and cannot be established directly from the effective theory.

2.12.2 Approach 2

Alternatively, in case we assume that all $k \in \mathcal{K}_t$ share the same black hole interior state such that:

$$|\psi'_{k,t}\rangle = |\psi'_{ext,k,t}\rangle |\chi\rangle \quad (13)$$

then this becomes a model where no Hilbert space factorization breakdown is required to guarantee the final black hole exterior purity, as long as $\sum_k |\beta_k|^2$ continues to increase monotonically over time up to 1. In this view, all black holes evaporate to the same final interior state, and this

is what grants the final exterior purity. The Page curve again becomes the theory determining $\mathcal{K}_t$.

In order for this process to be truly unitary, the transition process needs to be something like a generalized swap operator. This further assumes that right before transitioning to states with $k \in \mathcal{K}_t$, the black hole exterior horizon maintains a usual vacuum $|0_{ext,horizon}\rangle$ such that the transition process basically does the swap (gate) operation:

$$|0_{ext,horizon}\rangle|\psi_{int}\rangle \rightarrow |\psi_{int\rightarrow ext}\rangle|\chi\rangle \quad (14)$$

where $|\chi\rangle$ would then play a role of $|0\rangle$ and $|\psi_{int\rightarrow ext}\rangle$ is simply $|\psi_{int}\rangle$ cast around the black hole exterior part of the black hole horizon.

The swap operation can be interpreted more naturally - it represents the idea that while the semiclassical spacetime has the black hole exterior/interior factorization, there is no such factorization for branches $k \in \mathcal{K}_t$, and there is only black hole exterior. It can also be understood in terms of the islands proposal - instead of isolated islands, it is the **entire black hole interior** that disappears within a spacetime branch carrying $k \in \mathcal{K}_t$. Meanwhile, other branches $k \notin \mathcal{K}_t$ continue to have the black hole interior as the effective field theory predicts. Unlike the case where there is no superposition of spacetime and no energy conservation is imposed, the swap operation does not result in a rapid decrease in the entanglement entropy.

This alternative in the effective theory can therefore be summarized as: every spacetime branch within a spacetime superposition follows the usual effective field theory until when a branch (having radiated sufficient energy to the black hole exterior) transitions to states with $k \in \mathcal{K}_t$, where the final transition involves a generalized swap (gate) operation around the black hole horizon before there is no longer a horizon.

2.13 Digression: Page theorem and quantum measurements under black hole interior inaccessibility

The Page theorem also suggests the following: around the Page time, the universe with a black hole is approximately in a maximally mixed state, with the maximal ignorance about the black hole interior unless we already know the initial universe state.

If we further assume that the black hole interior becomes practically completely (that is, observationally and measurement-wise completely) inaccessible around the Page time (and there-

fore inaccessible in the effective theory) and the effective approximation can only be done to states with the support purely on the black hole exterior, then the effective black hole exterior state around the Page time is pure but can be arrived from different initial black hole configurations.

More practically stated, we may assume that from the Page time, observers in the black hole exterior have no means of deducing the black hole interior, including entanglement, in the effective theory such that the black hole exterior is effectively a pure state. All they can observe is the pure-state corrections from quantum gravity toward the effective theory, which we assume to eventually restore the full information in the black hole exterior thereby allowing for a unitary S-matrix despite intermediate-time non-unitarity.

This means that even if we started with the same effective-theory state, the final effective state cannot be deterministically predicted unitarily. Furthermore, there is maximal uncertainty over the initial fine-grained configurations that can lead to the same effective state.

The only prediction possible is a probabilistic prediction. The question is whether there is any effective-theory observable $\hat{O}_{eff}$ that commutes with the fine-grained evolution unitary such that it can serve as a pointer basis. In case the pointer basis exists, then we have the Zurek envariance setup of an equal-probability fine-grained state with the coarse-grained pointer basis, arriving at the Born rule [24].

Write the final ‘epistemic’ fine-grained quantum-gravity black hole exterior state as:

$$\rho_{epis,fine}^{final} = \frac{1}{N} \sum_{i=1}^N |\phi_i\rangle\langle\phi_i| \quad (15)$$

This reflects maximal classical uncertainty over initial fine-grained configurations, which we may purify as:

$$|\Psi\rangle = \frac{1}{\sqrt{N}} \sum_{i=1}^N e^{i\varphi_i} |\phi_i\rangle |e_i\rangle \quad (16)$$

In [24], the envariance argument then says probability $p(\phi_i) = 1/N$ for every i . Now let the pointer basis $\{|s_k\rangle\}_{k=1}^K$ of the effective-theory observable $\hat{O}_{eff}$ partition these N equally-weighted fine-grained branches, with N_k branches mapping to outcome k with $\sum_k N_k = N$. Since every fine-grained branch already carries equal weight $1/N$, coarse-graining back to the pointer basis assigns outcome k the probability

$$p(s_k) = \frac{N_k}{N}, \quad (17)$$

Because the pointer basis preserves effective amplitude information, we can refer to the effective state $|\psi\rangle$:

$$|\psi\rangle = \sum_k a_k |s_k\rangle \quad (18)$$

The final $|\Psi\rangle$ must preserve information $|a_k|^2$ via the pointer basis argument. Let $|\phi_i\rangle$ consistent with $|s_k\rangle$ be labeled with $|\phi_{\mu_{k,j}}\rangle$ (which has $i = \mu_{k,j}$), with $j = 1, \dots, N_k$. Therefore define:

$$\frac{1}{\sqrt{N}} \sum_{j=1}^{N_k} e^{i\varphi_{\mu_{k,j}}} |\phi_{\mu_{k,j}}\rangle |e_{\mu_{k,j}}\rangle \equiv e^{i\theta_k} \sqrt{|a_k|^2} |S_k\rangle \quad (19)$$

where $|S_k\rangle$ is a coarse-grained epistemic pointer basis that subsumes $|s_k\rangle$ as $|S_k\rangle \equiv |s_k\rangle |\varepsilon_k\rangle$, and there is no need to define $|\varepsilon_k\rangle$ precisely. This leads to:

$$|a_k|^2 = \frac{N_k}{N} = p(s_k) \quad (20)$$

We mention this measurement issue because of the approaches denying the Hilbert space factorization into the black hole interior/exterior or those allowing for a non-unitary effective theory despite a unitary S-matrix. In such a case, the conventional interpretation of the Born rule is no longer applicable - this was noted in the discussions after the AMPS firewall paradox was put forward [25] - but the damage to quantum mechanics may actually be more limited than often assumed, with the Born rule derived with classical maximal uncertainty over initial fine-grained states. That we can derive the Born rule also suggests that the non-unitary effective-theory approach can be consistent with a unitary S-matrix.

In relating to the typical quantum foundation debates, we note that this measurement theory comes from a **non-unitary** effective theory (or even if an effective theory is unitary, it is an ensemble theory) that emerges from a unitary quantum gravity theory. Therefore, the constraints that typically bind a hidden-variable measurement theory - in particular a local hidden-variable measurement theory - are inapplicable.

2.14 What is the black hole information problem then?

The effective theory needs to be addressed to resolve the black hole information problem completely, because the unitarity of a quantum gravity theory alone does not address what needs to be corrected in the effective theory to avoid the Page-time paradox, if we follow the Wallace argument that the Page-time paradox remains valid even when accepting the BER argument

regarding the generic non-unitarity of Cauchy-to-non-Cauchy transitions.

The Page-time paradox is mainly about the conflict between the thermodynamic description (based on the black hole horizon area rule and is supposedly the worst and simplest description) providing the upper bound to entanglement entropy and the effective field theory violating such an upper bound after the Page time. If we assume that there can only be small corrections in entanglement at each time from quantum gravity, then the unitarity of the effective theory implies that the upper bound remains significantly violated at late evaporation - this is the spirit of the small corrections theorem. A quantum gravity theory either addresses this issue or has some unknown inconsistencies or misunderstandings.

A quantum gravity theory can only address this issue by giving up one of the three - the unitarity of the effective theory, the no-drama equivalence principle ('small corrections') or effective Hilbert space factorization into the black hole interior/exterior. The unitarity violation destroys the assumption required for the validity of the small corrections theorem and its generalizations, the Hilbert space factorization breakdown is somewhat foreshadowed by the islands proposal, the no-drama equivalence principle violation is already widely discussed as heterodox options (such as 'fuzzballs'). And each breakdown does not have to be wholesale at each time and can be partial in a quantitative way, which is the point of Section 2.12.

Section 2.12 reveals the heavily conceptual nature of the black hole information problem. The conventional treatment of the problem is very unclear on how the paradox is resolved at the effective level, and section 2.12 shows how this may be possible despite the small corrections theorem and its generalizations without the wholesale (or whole-scale, if that word is to be preferred) sacrifice of the equivalence principle or unitarity in the effective theory.

In case the effective-theory black hole information problem is resolved, this tosses the question back to quantum gravity, which now needs to address the following conceptual questions:

1. Does quantum gravity really support the idea that the Hilbert space factorization breaks down (at least for some effective spacetime branches)? Is the breakdown limited to the effective theory, or is it about quantum gravity? This idea makes the Page-time paradox somewhat trivial, as there is no entanglement entropy to discuss anyway. So far, the answer to this question has been unclear in the mainstream approaches.
2. Does quantum gravity really support the idea that the effective theory can be non-unitary, if we keep the effective Hilbert space factorization? The Cauchy-to-non-Cauchy transitions

having non-unitarity evolutions generically seems to support the non-unitarity idea, with quantum gravity only providing corrections to restore the unitary S-matrix such that the Page-time paradox can be resolved. However, this seems to suggest that around black holes, many conventional assumptions and postulates about quantum mechanics are violated, including quantum measurements. Can this really be sustainable?

3. Alternatively, if we tweak the notion of ‘small corrections’ and the equivalence principle, we can be led to the alternative discussed in Section 2.12 in light of effective quantum gravity (‘a superposition of spacetime branches’). The question then becomes quantum-gravity verifications.

We now go into the generalization of the small corrections theorem in Section 3.

3 A generalization of the small corrections theorem

3.1 Small corrections generalization theorem

Effective-side unitarity + no-drama equivalence principle validates the semiclassical Hawking results for black hole evaporation. Assume that the full corrected effective

theory is unitary (that is, $U_{eff,t}$ is always unitary) then:

All states $|\psi\rangle \in \mathcal{H}_{eff}$ ($|\psi\rangle, |\phi\rangle$, etc.),

t refers to exterior observer time,

Δt being evolution time interval used in unitary $U_{eff,t}, U_{EFT,t}$,

$U_{EFT,t}$ evolution leads to the semiclassical Hawking results,

T being the current exterior observer time,

$S(t)$ roughly being naive coarse-grained entropy, large at every t ,

$$S_{\min} \equiv \min_{0 \leq t \leq T} S(t),$$

Exact value/meaning of $S(t)$ is not important, other than it being sufficiently large,

$$|\alpha_{\psi_{EFT,t}}|^2 = 1 - e^{-S(t)} \approx 1,$$

$$U_{eff,t}|\psi_{EFT,t}\rangle = \alpha_{\psi_{EFT,t}}U_{EFT,t}|\psi_{EFT,t}\rangle + \sqrt{1 - |\alpha_{\psi_{EFT,t}}|^2} |\phi_{correction,t}\rangle \Rightarrow$$

$$U_{eff,t}|\psi_{EFT,t}\rangle \approx U_{EFT,t}|\psi_{EFT,t}\rangle \Rightarrow$$

$$U_{eff,T}U_{eff,T-\Delta t}..U_{eff,0}|\psi_{EFT,0}\rangle \approx U_{EFT,T}U_{EFT,T-\Delta t}..U_{EFT,0}|\psi_{EFT,0}\rangle \quad \text{for } T \ll e^{S_{\min}/2} \text{ (by unitarity)} \quad (21)$$

For Equation (21), we further clarify as follows:

1. $U_{eff,t}$ is allowed to be state-dependent and time-dependent from the start, so the result cannot be avoided by resorting to state-dependent maps, including state-dependent error-correcting codes, unless unitarity is further violated.
2. $U_{eff,t}$ works solely on $\mathcal{H}_{eff}$, the effective Hilbert space, so the objection that ‘quantum gravity states are structurally different from effective states’ (invoking the eigenstate thermalization hypothesis: ETH) cannot be used to avoid the result.
3. The EFT evolution operator $U_{EFT,t}$ - the quantum-gravity-uncorrected effective field theory around the semiclassical horizon - is allowed to be state-dependent and time-dependent, independently of $|\phi_{correction,t}\rangle$. Furthermore, while we would usually assume $U_{EFT,t}$ is unitary, it does not have to be unitary. This is a preemptive response to the objection that features such as the black hole singularity already force corrections to the naive effective field theory well before any quantum-gravity correction is considered such that no genuinely unitary, state-independent $U_{EFT,t}$ is feasible. Any such pre-quantum-gravity correction can simply be absorbed into $U_{EFT,t}$. This is also useful when handling

the unitarity question of replica wormholes.

4. $U_{EFT,t}$ can incorporate large corrections of the original effective theory that are irrelevant to radiations around the black hole horizon.
5. We assume that EFT maintains the usual pair entanglement around the black horizon that drives the monotonically increasing entanglement entropy (‘Hawking result’).
6. While we mainly use Equation (21) for addressing the entanglement entropy question, it can be applied even when the Hilbert space factorization into the black hole interior/exterior breaks down. In such a case, the entropy question cannot be defined, but the general question of how far the corrections can accumulate from small corrections of the original EFT can be addressed.
7. Quantum gravity’s role in Equation (21) can only be about quantum gravity states providing effective-theory or effective-state corrections. After corrections, quantum gravity is completely irrelevant. Therefore, the fact that states of the full quantum gravity theory are expected to differ significantly from effective-theory states is irrelevant to the results.

Translated to words from Equation (21): **even if** the unitary effective theory is **state-dependent and time-dependent** around the black hole horizon (which means ‘unitary theory’ itself may be state-dependent such that unitary $U_{eff,t}$ may not be the same for different current effective states), especially at late-time black hole evaporation, as long as the Δt -corrections around the conventional semiclassical EFT-like state remain exponentially small at every time within the effective theory (no-drama **equivalence principle** / small corrections), the effective-side deviations of the black hole from the original effective field theory are guaranteed to remain **exponentially small** throughout the entire black hole evaporation.

This means that in order to address the Page-time paradox, we need to give up at least one of the three:

1. Approximate no-drama equivalence principle,
2. Hilbert space factorization into the black hole exterior/interior,
3. Unitarity of the effective theory.

3.2 The original small corrections theorem

For simplicity but without the loss of generality, Hawking radiation can be pictured as a maximally entangled qubit pair of an outgoing qubit and its infalling qubit partner, which the Mathur model adopts [3]:

$$|\Psi\rangle_{pair} = \frac{1}{\sqrt{2}}|0\rangle_c|0\rangle_b + \frac{1}{\sqrt{2}}|1\rangle_c|1\rangle_b, \quad (22)$$

This is a significant simplification of the thermal state but is nevertheless basically valid for the entanglement entropy purpose. After N such emissions the radiation quanta $\{b_i\}$ are entangled with the infalling partners and the infalling matter with entanglement entropy $S_{ent} = N \ln 2$, growing without bound until the final evaporation.

Mathur's small corrections theorem asks whether this monotonic increase in S_{ent} can be avoided once the leading-order pair state at each emission is allowed to deviate from $|\Psi\rangle_{pair}$ in small amount ϵ (we will not define this ϵ exactly, leaving its detailed expositions to [3]). Using only the subadditivity and strong subadditivity of entanglement entropy, Mathur proves that the entanglement entropy of the emitted radiation must increase by at least $\ln 2 - 2\epsilon$ at every single emission, so that after N emissions $S_{ent} > N(\ln 2 - 2\epsilon)$. Since restoring the black hole exterior purity requires this growth to eventually halt and reverse, that is only possible if $\epsilon = O(1)$.

Equation (21) allows for more general types of small corrections, and suggest that there is not much we can do once we assume the unitarity of the effective theory. Equation (21) also does not require the horizon emission structure of the Mathur's small corrections theorem. All it says is that if we want large corrections to (unspecified) effective theory predictions around the black hole horizon, then they cannot arise from accumulation of exponentially small corrections over non-exponential time-scale unless we give up something.

3.3 Remaining derivations of Equation (21)

We now derive the last equation of Equation (21)

$$U_{eff,T}U_{eff,T-\Delta t}..U_{eff,0}|\psi_{EFT,0}\rangle \approx U_{EFT,T}U_{EFT,T-\Delta t}..U_{EFT,0}|\psi_{EFT,0}\rangle \quad (23)$$

from

$$U_{eff,t}|\psi_{EFT,t}\rangle = \alpha_{\psi_{EFT,t}}U_{EFT,t}|\psi_{EFT,t}\rangle + \sqrt{1 - |\alpha_{\psi_{EFT,t}}|^2}|\phi_{correction,t}\rangle \quad (24)$$

Take $U_{EFT,t}$ to propagate the EFT-like state family forward, $U_{EFT,t}|\psi_{EFT,t}\rangle = |\psi_{EFT,t+\Delta t}\rangle$, and abbreviate $\alpha_t \equiv \alpha_{\psi_{EFT,t}}$, $S_k \equiv S(k\Delta t)$, $|\Psi_k\rangle \equiv U_{eff,k\Delta t} \cdots U_{eff,0}|\psi_{EFT,0}\rangle$, $|\Phi_k\rangle \equiv U_{EFT,k\Delta t} \cdots U_{EFT,0}|\psi_{EFT,0}\rangle = |\psi_{EFT,(k+1)\Delta t}\rangle$, and $\epsilon_k \equiv |||\Psi_k\rangle - |\Phi_k\rangle||$. $S(t)$ need not be constant in time; only $S(t)$ large at every $t \in [0, T]$ is used below.

Write $|\Psi_{k-1}\rangle = |\Phi_{k-1}\rangle + |\delta_{k-1}\rangle$ with $||\delta_{k-1}\rangle|| = \epsilon_{k-1}$. Since $|\Phi_{k-1}\rangle = |\psi_{EFT,k\Delta t}\rangle$,

$$U_{eff,k\Delta t}|\Phi_{k-1}\rangle = \alpha_k|\Phi_k\rangle + \sqrt{1 - |\alpha_k|^2}|\phi_{correction,k\Delta t}\rangle, \quad (25)$$

and unitarity of $U_{eff,k\Delta t}$ preserves the norm of the second piece, $||U_{eff,k\Delta t}|\delta_{k-1}\rangle|| = \epsilon_{k-1}$. So

$$|\Psi_k\rangle - |\Phi_k\rangle = (\alpha_k - 1)|\Phi_k\rangle + \sqrt{1 - |\alpha_k|^2}|\phi_{correction,k\Delta t}\rangle + U_{eff,k\Delta t}|\delta_{k-1}\rangle, \quad (26)$$

and, by the triangle inequality,

$$\epsilon_k \leq |\alpha_k - 1| + \sqrt{1 - |\alpha_k|^2} + \epsilon_{k-1}. \quad (27)$$

With $|\alpha_k|^2 = 1 - e^{-S_k}$ (taking α_k real and positive without loss of generality), $\sqrt{1 - |\alpha_k|^2} = e^{-S_k/2}$ and $|\alpha_k - 1| = 1 - \sqrt{1 - e^{-S_k}} \approx \frac{1}{2}e^{-S_k} \ll e^{-S_k/2}$ for $S_k \gg 1$, so to leading order

$$\epsilon_k \lesssim \epsilon_{k-1} + e^{-S_k/2}. \quad (28)$$

Since $\epsilon_{-1} = 0$, induction gives

$$\epsilon_T \lesssim \sum_{k=0}^T e^{-S_k/2} \leq (T+1)e^{-S_{\min}/2}, \quad S_{\min} \equiv \min_{0 \leq k \leq T} S_k, \quad (29)$$

i.e.

$$||U_{eff,T} \cdots U_{eff,0}|\psi_{EFT,0}\rangle - U_{EFT,T}U_{EFT,T-\Delta t} \cdots U_{EFT,0}|\psi_{EFT,0}\rangle|| \lesssim (T+1)e^{-S_{\min}/2}, \quad (30)$$

which is $\ll 1$ precisely for $T \ll e^{S_{\min}/2}$, recovering the claimed approximate equality.

4 Conclusion

This paper revisits the question of what the black hole information problem is actually about, in light of recent progress in mainstream quantum gravity physics. The key question becomes:

is the black hole information problem the intermediate-time paradox toward the final purity of the black hole exterior in the **full quantum gravity theory**, or the final purity in the **effective theory**?

Section 2 surveyed how unsettled this question is. BER's algebraic argument [1] concludes that ordinary evaporation evolves a pure state to a mixed one, and traces the so-called paradox to confusion regarding Cauchy-to-non-Cauchy transitions on the spacetime that is not globally hyperbolic. Wallace [2] argues that the black hole information paradox actually considered in quantum gravity is not just about final purity but the Page-time paradox - the intermediate-time conflict between the horizon area rule-based thermodynamic entanglement entropy upper bound and the actual entanglement entropy calculation.

Assuming the Page-time paradox is actually valid, we asked what the paradox actually refers to - is it about the effective theory that our usual observables live in or the full quantum gravity theory? The paradox is naturally stated using effective-theory quantities - the area rule and the semiclassical entropy calculation both make sense without reference to any specific full quantum gravity theory, and any resolution should be able to state what went wrong in the effective theory calculations.

And yet the mainstream physics literature often treats purity in the quantum gravity theory as settling the matter. We showed the situation is more subtle. That the full quantum theory has the final black hole exterior purity cannot imply that the effective theory restores the final pure black hole exterior state, once the effective theory is constrained by the equivalence principle and unitarity. This violates the thermodynamic upper bound set by the horizon area rule (Bekenstein-Hawking entropy), which implies that the main black hole information problem should first be cast in the effective theory, not in the full quantum gravity theory.

Even when the effective theory has the final black hole exterior purity, the full quantum gravity theory is not guaranteed to have the final pure black hole exterior state. In this case, the main black hole information problem shifts to the full quantum gravity theory, since the fine-grained entanglement entropy in the full quantum gravity theory violates the thermodynamic upper bound.

All of these would not be serious issues if some unitary effective theory were consistent with final black hole exterior purity in both the effective theory and the full quantum gravity theory without significantly violating the equivalence principle. Equation (21) and the small corrections theorem [3] together show this is impossible. At least one of the equivalence principle,

effective bulk Hilbert space factorization, or unitarity of the effective theory has to be given up, and mainstream approaches typically sacrifice effective-theory unitarity without explicit acknowledgment. But practical uses of quantum mechanics, including quantum computation and quantum measurements, assume effective-theory unitarity. Whether new physics such as non-linear effective theory or a new theory of quantum measurements is acceptable remains unaddressed by the mainstream approaches.

What this leaves is a state of conceptual unclarity in quantum black hole physics. Even granting the mainstream physics view, it remains unclear what showing ‘information is preserved’ in the black hole information problem is actually supposed to establish. Is effective-theory unitarity treated as a real constraint, or is it tacitly abandoned on the assumption that only full-theory unitarity needs to be shown? How much of the equivalence principle or the Hilbert space factorization is quantum gravity willing to sacrifice? Mainstream treatments of the black hole information problem have, so far, offered muddled answers to these questions. This paper has taken steps for clarification, and we wish that more steps toward conceptual clarity are taken in mainstream quantum gravity physics.

Data availability and declaration of interests

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